PCBC H Parametric & Sequences Test (Practice A) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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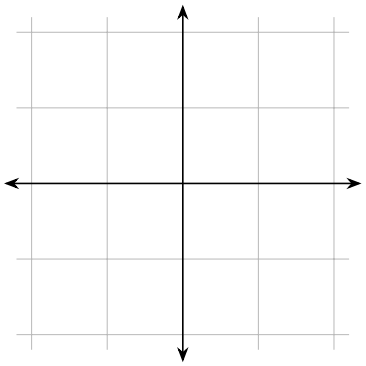
_____ _____

Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

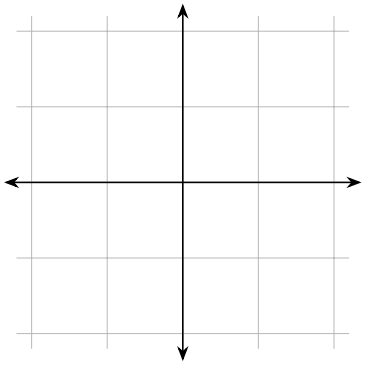
1. $x = 2t + 2, \quad y = t^2 + 2t + 1, \quad -3 \leq t \leq 1$

SIMPLIFIED rectangular form: _____



2. $x = \sqrt{t}, \quad y = 2t - 4, \quad 0 \leq t \leq 4$

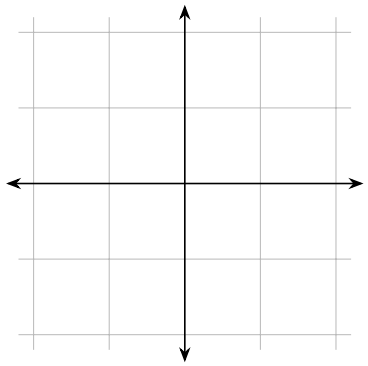
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

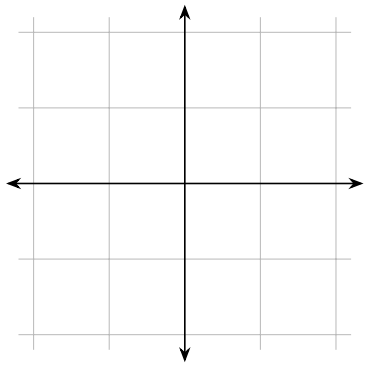
3. $x = 3\cos t, \quad y = 5\sin t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = 2\sec t, \quad y = 3\tan t$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = -3(x+2)^2 + 5$

Express the sum using summation notation. (2 points)

6. $\frac{1}{3} + \frac{2}{4} + \frac{3}{5} + \cdots + \frac{15}{17}$

Extra practice: the other sequence questions #6 could be

Not part of the 16 points. One of these types is the one sequence question; worked in the key.

Write down the first five terms of the sequence.

E1. $\{a_n\} = \left\{ \dfrac{2^n}{n^2+1} \right\}$

A sequence is defined recursively. Write the first five terms.

E2. $a_1 = 6; \quad a_n = \dfrac{a_{n-1}}{n}$

Write out the sum.

E3. $\displaystyle\sum_{k=1}^{n} (2k+5)$

Answer Key

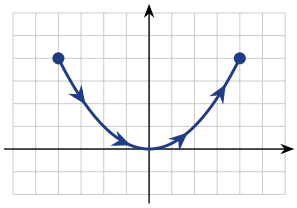
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = 2t+2,\ y = t^2+2t+1,\ -3 \le t \le 1$

t	x	y
-3	-4	4
-2	-2	1
-1	0	0
0	2	1
1	4	4

brackets: plug in both endpoints



solid dots, no end arrows: closed interval

$x-2 = 2t \ \Rightarrow\ t = \frac{x-2}{2}$ — *isolate t in x: one answer, no $\pm$*

$y = \left(\frac{x-2}{2}\right)^2 + 2\left(\frac{x-2}{2}\right) + 1$ — *substitute*

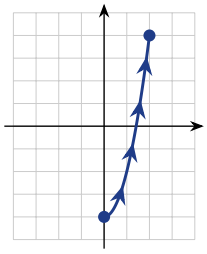
$y = \frac{x^2-4x+4}{4} + x - 2 + 1$

$\boxed{y = \frac{1}{4}x^2 \ \text{ for } x \text{ in } [-4,4]}$ — *SIMPLIFIED = y alone; domain = x from the t-chart*

Other way: $y = (t+1)^2$ and $t+1 = \frac{x}{2}$, so $y = \frac{x^2}{4}$.

2. $x = \sqrt{t},\ y = 2t-4,\ 0 \le t \le 4$

t	x	y
0	0	-4
1	1	-2
4	2	4



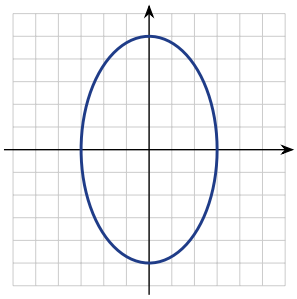
$x^2 = t$ — *square both sides: one answer for t*

$y = 2x^2 - 4$ — *substitute*

$\boxed{y = 2x^2 - 4 \ \text{ for } x \text{ in } [0,2]}$ — *$x = \sqrt{t}$ is never negative, so only the right half*

3. $x = 3\cos t,\ y = 5\sin t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(3,0)$
$\frac{\pi}{2}$	$(0,5)$
π	$(-3,0)$
$\frac{3\pi}{2}$	$(0,-5)$
2π	$(3,0)$



direction not asked here (it would be counterclockwise from $(3,0)$)

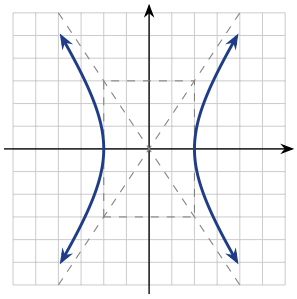
$\cos t = \frac{x}{3}, \quad \sin t = \frac{y}{5}$ — *solve each for the trig function*

$\sin^2 t + \cos^2 t = 1 \Rightarrow \left(\frac{y}{5}\right)^2 + \left(\frac{x}{3}\right)^2 = 1$ — *identity*

$\boxed{\frac{x^2}{9} + \frac{y^2}{25} = 1}$ — *whole ellipse is traced: no restriction*

4. $x = 2\sec t,\ y = 3\tan t$ (no interval given)

t	(x,y)
$-\frac{\pi}{4}$	$(2\sqrt{2},-3)$
0	$(2,0)$
$\frac{\pi}{4}$	$(2\sqrt{2},3)$
$\frac{3\pi}{4}$	$(-2\sqrt{2},-3)$
π	$(-2,0)$
$\frac{5\pi}{4}$	$(-2\sqrt{2},3)$



box from $a=2$, $b=3$; asymptotes through the corners; end arrowheads: both branches keep going

$\sec t = \frac{x}{2}, \quad \tan t = \frac{y}{3}$

$1+\tan^2 t = \sec^2 t \Rightarrow 1+\frac{y^2}{9} = \frac{x^2}{4}$ *identity*

$\boxed{\frac{x^2}{4} - \frac{y^2}{9} = 1}$ *t-chart: $(2,0)$ and $(-2,0)$, both branches: whole graph, done*

also right: $y = \pm 3\sqrt{\frac{x^2}{4}-1}$ *keep $\pm$; $+\sqrt{}$ alone is wrong*

5. $y = -3(x+2)^2+5$

$x = t,\ y = -3(t+2)^2+5$ *replace x with t*

$\boxed{x = t,\ y = -3(t+2)^2+5,\ t \text{ in } (-\infty,\infty)}$

$x = t-2:\ y = -3(t-2+2)^2+5$ *pick $t-2$ so the inside becomes t*

$\boxed{x = t-2,\ y = -3t^2+5,\ t \text{ in } (-\infty,\infty)}$

Give both $x =$ and $y =$ ($t = x+2$ is not parametric form). "For t in $(-\infty,\infty)$" is assumed for a whole parabola; writing it is fine. Don't expand $-3(t+2)^2+5$; it's not asked and it wastes time.

6. $\frac{1}{3}^{k=1} + \frac{2}{4}^{k=2} + \frac{3}{5}^{k=3} + \cdots + \frac{15}{17}^{k=15}$

top: k, bottom: $k+2$ *top counts 1 to 15; bottom is always 2 more*

$\boxed{\sum_{k=1}^{15} \frac{k}{k+2}}$ *lower index 1, upper index 15, same letter everywhere*

Check: $k=1$ gives $\frac{1}{3}$ and $k=15$ gives $\frac{15}{17}$.

Extra practice (not scored)

E1. $\left\{\frac{2^n}{n^2+1}\right\}$

$\frac{2}{2}, \frac{4}{5}, \frac{8}{10}, \frac{16}{17}, \frac{32}{26}$ *plug in $n=1$ to 5: bottoms 2, 5, 10, 17, 26*

$\boxed{1, \frac{4}{5}, \frac{4}{5}, \frac{16}{17}, \frac{16}{13}}$ *reduce: $\frac{2}{2}=1$, $\frac{8}{10}=\frac{4}{5}$, $\frac{32}{26}=\frac{16}{13}$*

E2. $a_1 = 6,\ a_n = \frac{a_{n-1}}{n}$

$a_2 = \frac{6}{2} = 3, \;\; a_3 = \frac{3}{3} = 1$ *divide the previous term by n*

$a_4 = \frac{1}{4}, \;\; a_5 = \frac{1/4}{5} = \frac{1}{20}$

$\boxed{6,\ 3,\ 1,\ \frac{1}{4},\ \frac{1}{20}}$

E3. $\sum_{k=1}^{n}(2k+5)$

$(2\cdot 1+5)+(2\cdot 2+5)+(2\cdot 3+5)+\cdots+(2n+5)$ *plug in $k = 1,2,3$, then the last term with n*

$\boxed{7+9+11+\cdots+(2n+5)}$ *don't add*